

1. Complete the table by writing the constraint given as a written description using an inequality and set-builder notation.

Written Description	Inequality	Set-Builder Notation
All numbers greater than or equal to negative eight		

2. A table of values that represents a function is shown.

x	-4	-1.25	0	1	3	6.25	10
y	8.2	6	5	4.2	2.6	0	-3

Use the table of values to determine the indicated key features for the function that is represented.

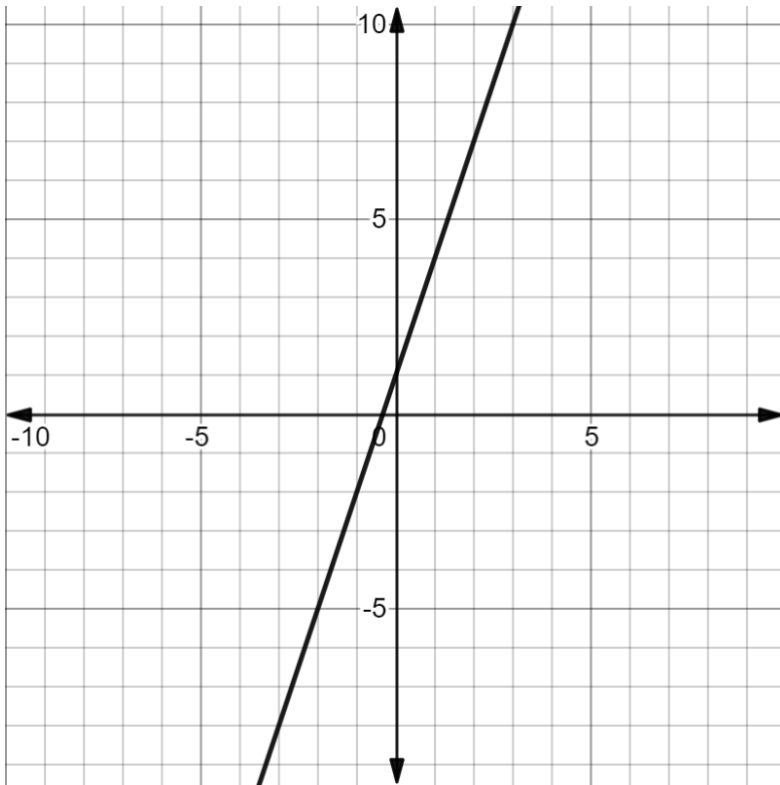
Key Feature	End Behavior	y -intercept
Description	As x -values increase, y -values _____.	The y -intercept is _____.

3. Write the equation of the line that goes through the point $(4, -7)$ and is perpendicular to the line $y + 6 = -\frac{2}{5}(x - 1)$. Use the table to show your work, if needed, to determine each form, and then give each form.

Work for Point-Slope Form	Work for Slope-Intercept Form
Point-Slope Form	Slope-Intercept Form

4. Write the equation of the line that is parallel to $x = 4$ and passes through $(-3, 7)$.

5. Write the equation of the line in standard form that passes through $(4, 2)$ and is parallel to the line shown on the coordinate grid.



Show your work.

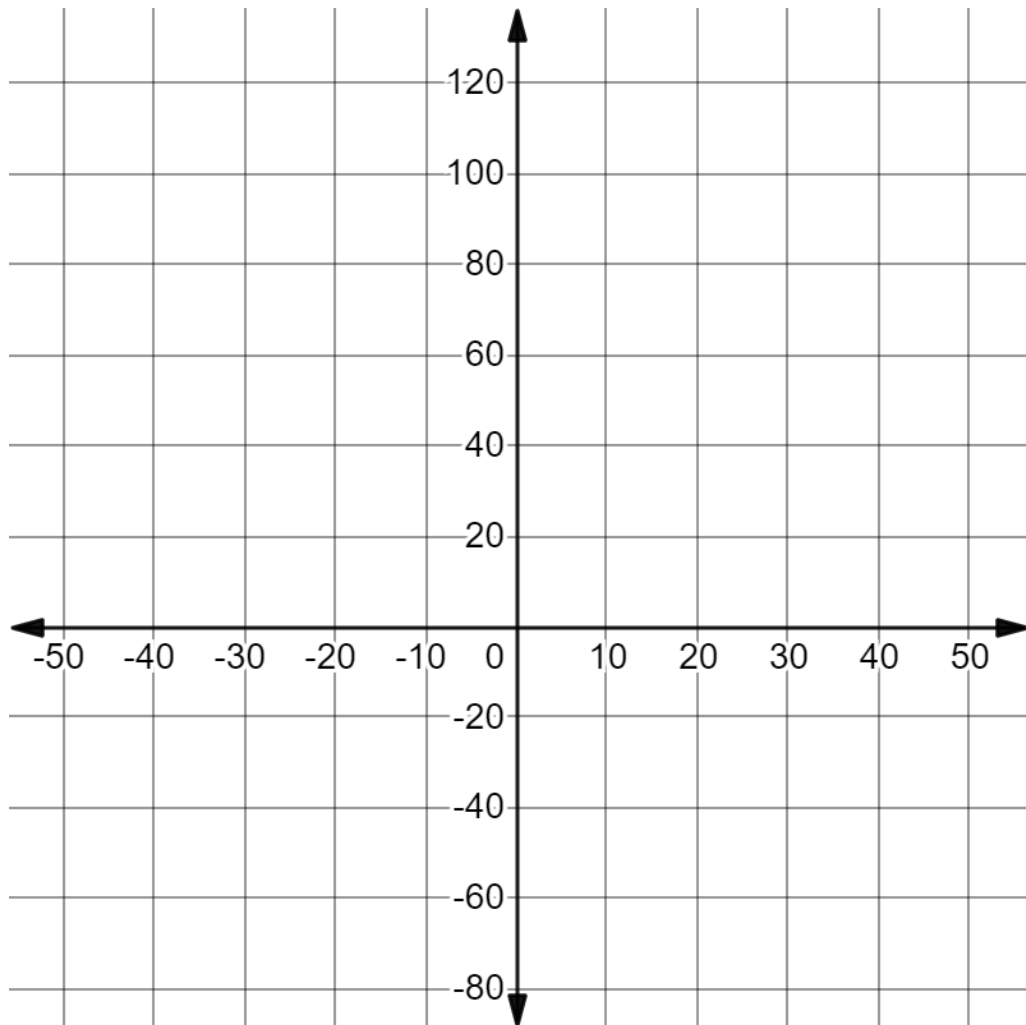
Standard form of parallel line _____

6. A research firm determined that the equation $y = 2x + 80$ can be used to model the total amount of spending by tourists in Florida, y , in billions of dollars, for the years 2013-2017 where x is the number of years since 2013.

a. Determine the key features for the context. Show any necessary work below the table.

Domain	Range	As x -values increase, y -values ...
Slope	x -intercept	y -intercept

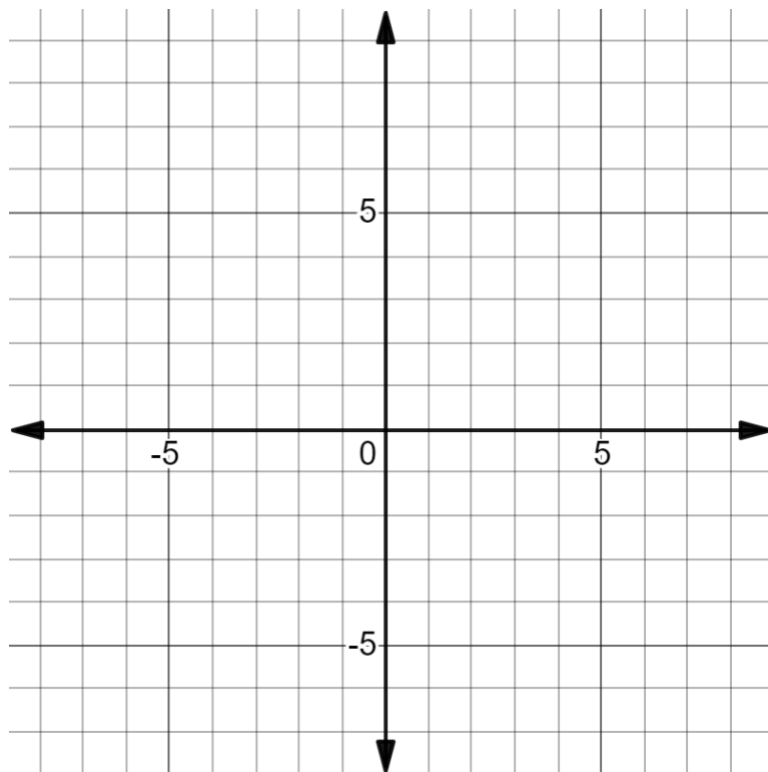
b. Graph the equation $y = 2x + 80$.



7. Graph the function $f(x) = -\frac{5}{4}x + 6$.

Label the point on the graph that is the solution to $f(x) = -4$ by writing "solution" next to it. Then, write the solution below.

Solution



8. The function $S(t) = 1.084t + 17.392$ models the amount of money, S , in billions of dollars, the United States government spent on science, space, and technology for the time period 1999-2009, where t is the number of years since 1999.

a. Write the constraints for the variable t using set-builder notation.

Constraints _____

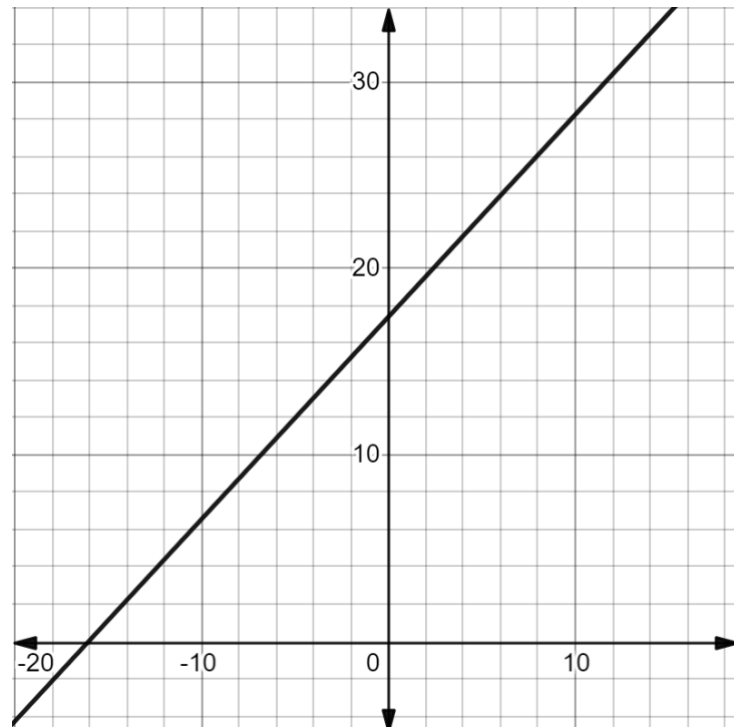
b. Determine the value of $S(7)$.

c. Interpret the value of $S(7)$ in context.

d. Name the key feature that is found when solving $S(t) = 0$. Then, label the key feature on the graph of $S(t) = 1.084t + 17.392$.

Key Feature _____

e. Determine if the value for $S(t) = 0$ is reasonable for the context. Explain why or why not.

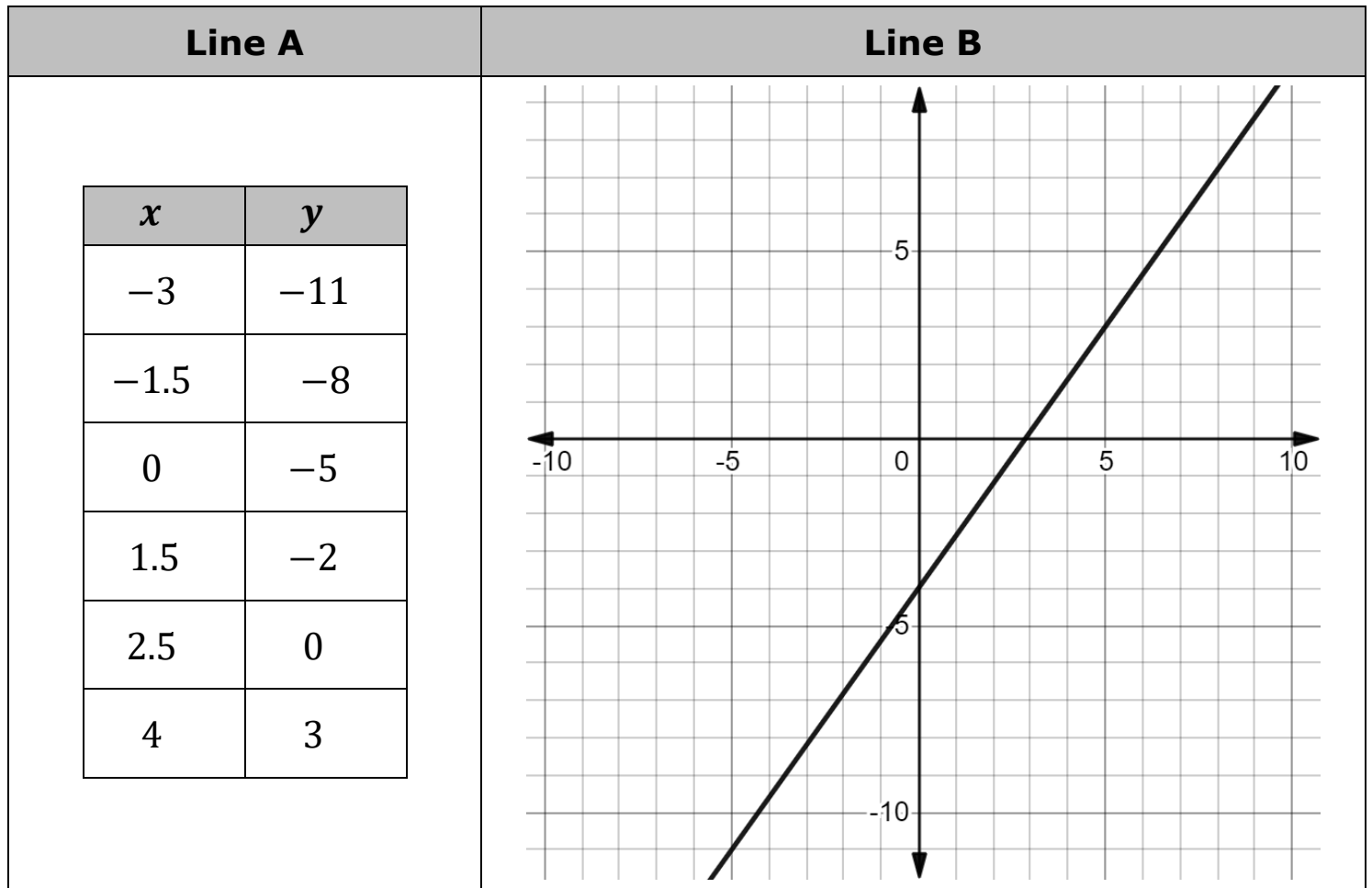


Reasonable?

- ☐ No
☐ Yes

Explanation _____

9. Two different linear functions are shown.



Select the phrases that complete the comparison of the key features.

The x -intercept for

- ☐ line A
- ☐ line B

is greater than the x -intercept for

- ☐ line A.
- ☐ line B.

The range for line A is

- ☐ the same as
- ☐ different than

the range for line B.

The slope for

- ☐ line A
- ☐ line B

is smaller than the slope for

- ☐ line A.
- ☐ line B.